Data Denoising and Post-Denoising Corrections in Single Cell RNA Sequencing

Applied Bayesian Reading Group

Suppapat Korsurat
skorsu@umich.edu

July 9, 2020

Outline

- Introduction
  - Single Cell RNA Sequencing
  - Biological Noise
- Technical Noise in Single Cell RNA Sequencing
- Denoising Single Cell Data
- Post-Denoising Analysis and Inference
  - Estimating Function of X
  - Gene Expression Differential Testing
- Reference

Single Cell RNA Sequencing (scRNA-seq)

- The technology uses in exploring the gene expression profile at the single-cell level.
- The advantages of using scRNA-seq
  - Bulk sequencing requires a large number of cells. Sometimes, it will be difficult to find enough cells due to its rarity.
  - In scRNA-seq, each cell will be represented by the entire transcriptome.
    - the discovery of new subpopulations of cells.
    - the deeper understanding of what constitutes specific cell types, how cell types function, and how they may change during disease.

Biological Noise

- Noise is a variation between cells of the same type.
- Gene Expression Noise is a stochastic variation in RNA count between cells of the same overt type.
- There are many reasons that causing the Gene Expression Noise such as cell size, cell location within tissue, fluctuations in the expression of upstream genes.
- Extrinsic Noise is a variation caused by the external factors, such as global, environmental factors.
  - latent factors that induce correlations between genes and cells.
- Intrinsic Noise is a variation owing to the stochasticity inherent in the biochemical process of gene expression.
  - fluctuations that are independent across genes and cells.

Experiment: RNA Extraction

Consider the process of scRNA-seq RNA.

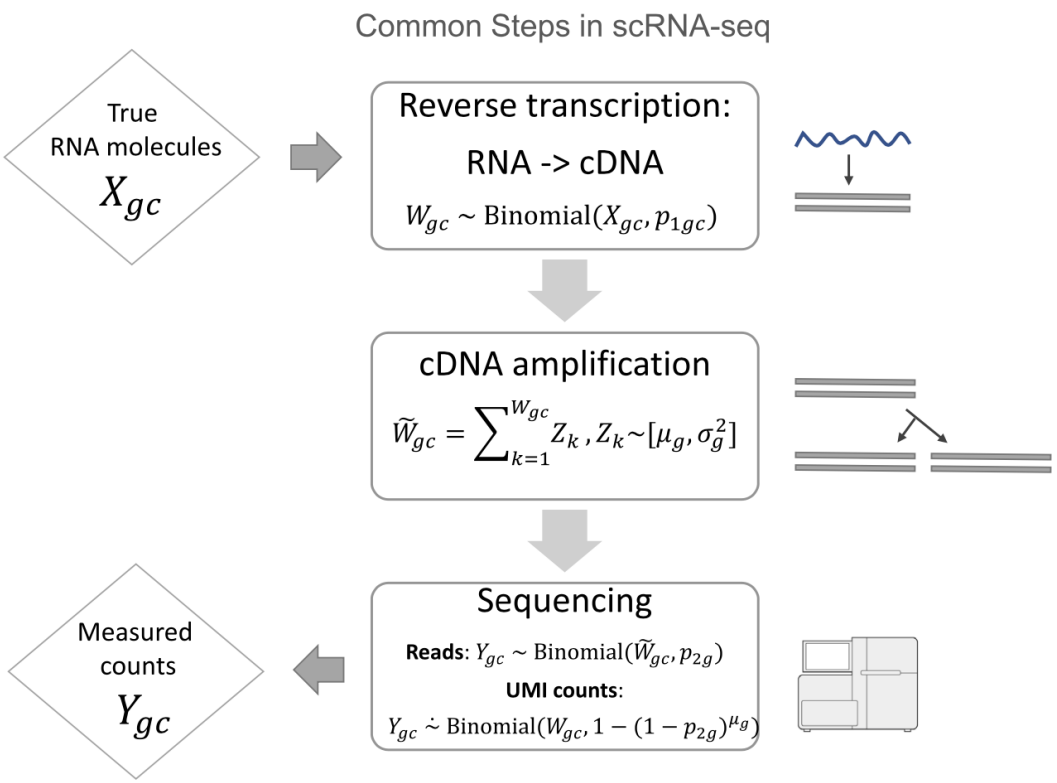

FIG. 2. *Propagation of measurement error during scRNA-seq a experiment. In the notation, subscript c indexes cells and g indexes genes. See the Supplementary Material* (Agarwal, Wang and Zhang, 2020) *for details.*

Experiment: RNA Extraction

The first step is called Reverse Transcription step.

- Reverse transcription is a technique for generating a complementary strand of DNA (cDNA) from RNA.
- Only a fraction of the RNA molecules are successfully converted into cDNA. (Dropout Event)

Experiment: Reverse Transcription

- Let X_{gc} is a number of RNA molecule of gene g and cell c
- Let W_{gc} is a cDNA of gene g and cell c.
- $W_{gc} \sim Binomial(X_{gc}, p_{1gc})$
  - p_{1gc} is an efficiency of the reverse transcription process of gene g and cell c.

Experiment: RNA Extraction

The second step is a cDNA Amplification step.

- In this step, we will increase the number of cDNA molecule.
- Barcode Strategy (UMI; Unique Molecular Quantifier)
  - Each cDNA molecule is tagged with a unique barcode before amplification.

Experiment: cDNA Amplification

- Let Z_k be the kth copy of gene g in cell c.
  The distribution of Z_k is unknown but the mean is μ_g and the variane is σ_g^2
- Let $\tilde{W}_{gc}$ is a total number of cDNA after the cDNA Amplification process.
  $\tilde{W}_{gc} = \sum_{k=1}^{W_{gc}} Z_k$

Experiment: RNA Extraction

The final step is a Sequencing step.

- In this step, the sequencer will read the amplified cDNA, the result from previous step, and map them to the annotated genome template.
- Since the sequencing has a low efficiency to map them back, the amplification step is required.

Experiment: RNA Extraction

- Let Y_{gc} is a measured count of RNA molecule of gene g and cell c.
- p_{2g} is an efficiency of the sequencing process of gene g and cell c. $Y_{gc} \sim Binomial(\tilde{W}_{gc}, p_{2g})$

Experiment: RNA Extraction

- The distribution stated in previous slide is an observed count without UMI.
- If we using this distribution, the distribution of $Y_{gc}|X_{gc}$ will be difficult to find due to the aggregated measurement error.
- UMI strategy: identify that the unique cDNA is sequenced before the amplification process or not.

Experiment: RNA Extraction

Experiment: RNA Extraction (with UMI)

- Assume that each of the kth copy has the same probability that at least one of the amplified copied is sequenced, $1-(1-p_{2g})^{\mu_g}$
  $Y_{gc} \sim Binomial(W_{gc}, 1-(1-p_{2g})^{\mu_g})$
- By using this, we can find that $Y_{gc}|X_{gc} \sim Binomial(X_{gc}, \alpha_{gc})$ where $\alpha_{gc} = p_{1gc}(1-(1-p_{2g})^{\mu_g})$
- α_{gc} can be considered as the overall efficiency.
  - p_{1g} and p_{2g} are the efficiency of the reverse transcription and sequencing process of gene g and cell c respectively.
- Most of the time, α_{gc} is less than 0.1 and X_{gc} is large. It is reasonable to approximate Binomial distribution with the Poisson Distribution. $Y_{gc}|X_{gc} \sim Poisson(X_{gc}\alpha_{gc})$

Model for the Technical Noise

- Even though the technical noise distribution of $Y_{gc}|X_{gc}$ spread across the three steps is hard to measure since it is impossible to have true technical replicates, we still need assumptions for the technical noise in order to denoise the data and estimate the underlying gene expression values.
- Since there are a lot of 0 in the measured counts, many methods has been proposed by using zero-inflated mixture models.
  - Difficult to fit (both statistical and computational efficiency aspects.)
  - To determine that the 0 is a true value or the technical noise, more assumptions and/or additional information are needed.

Model for the Technical Noise

- This problem is solved by using the UMI in the process.
  - Assume that the true number of RNA copies in gene g and cell c is X_{gc} and the observed count is Y_{gc}.
  - PCR amplification variation will be ignored.
  - $Y_{gc}|X_{gc}$ will be Poisson Distribution instead of Binomial Distribution.
  - Advatages
    - No need an extra zero inflation term for dropout.
    - All technical zeroes are modeled by. Poisson-based down-sampling.

Denoising

- In the scRNA-seq, Most of the entries in the cell-by-gene count matrix are zeros.
  - true zeros because of no expression. (not a missing value)
  - technical zero (exact missing value) due to
    - dropout during reverse transcription.
    - sampling during sequencing.
- We need to denoise it.
  - recover the original RNA counts for each cell and each gene.
- Denoising is different from the classical imputations.
- The objectives of doing data denoising.
  - to recover the genes that are missing.
  - to improve the estimates of all genes.

Method for Denoising scRNA-seq Data

- The key is to borrow the information across the similar cells/genes.
- There are a lot of methods for denoising the data.
  - Smoothing across neighboring cells that are proximal in lower-dimensional geometrical representation.
    - kNN-smoothing, VIPER, scImpute, Markov Affinity-based Graph Imputation of Cells (MAGIC)
  - Using the gene-gene correlations.
    - SAVER, DCA, scVI, ALRA

SAVER

- SAVER stands for Single-Cell Analysis via Expression Recovery

SAVER Model

$Y_{gc}|X_{gc} \sim Poisson(\alpha_{gc}X_{gc})$
$X_{gc} \sim Gamma(a_{gc}, b_{gc})$: a_{gc} is shape and b_{gc} is rate parameter.

- $E[X_{gc}] = \mu_x = \frac{a_{gc}}{b_{gc}}$ and $Var[X_{gc}] = \sigma^2_{gc} = \frac{a_{gc}}{b^2_{gc}}$
- By using Empirical-Bayes like technique, we can find the μ_x and σ^2_{gc}, which we will use them to calculate the parameters of the prior distribution.

- By the property of Gamma-Poisson Model, the Y_{gc} will be a Negative Binomial random variable.

SAVER

- However, the experiment (Figure 2 in the paper) introduces the technical noise in every steps. This will introduce a bias in the observed expression count. (Y_{gc})
- To eliminate this problem, we need
  - to distinguish the biological noise from the technical noise in the matrix Y.
  - a careful choice of technical noise model.

Biological Noise between cells

- Consider a general framework where the true gene expression is derived by adding independent stochastic noise to an underlying correlated component, Λ_{gc}

Equation 4.1

$$X_{gc}|\Lambda_{gc} \overset{ind}{\sim} F(\Lambda_{gc}, \varphi_g \Lambda_{gc})$$

- Λ_{gc} is the portion of gene g's expression that is predictable given the expression of other genes.
- F is a distribution for the intrinsic noise.
- The assumption of F is that the expected magnitude of the deviation of X_{gc} from Λ_{gc} is controlled by a single gene-specific dispersion parameter φ_g.

Estimating Function of X

- Let $\hat{X}$ be a result from SAVER model.
- Let $f(X)$ be a function of interest.
- If you plug in the $\hat{X}$ into the $f(X)$ directly, the result will be biased as the uncertainty of in the estimation of $\hat{X}$ is ignored.
- Strategies to eliminate this problem
  - Using the posterior mean, result from the SAVER framework, and plug in the function of interest. $E[f(X)|Y, \Lambda = \hat{\Lambda}]$
  - Sampling from the posterior distribution of X.
    - We will use this strategy if the analytical form of the function of interest is hard to derive.

Result of the SAVER

- SAVER framework will give the posterior distribution of X. Specifically, The posterior will be $X|Y, \Lambda$
- The estimates can be improved by using the posterior mean, $E[X|Y, \Lambda]$.
- However, Λ is unknown. Additionally, there is no strategies that can computed this term easily.
- We will use $\hat{\Lambda}$ as an estimator of Λ and plug it in the posterior mean, which is $E[X|Y, \Lambda = \hat{\Lambda}]$

Estimating Function of X: Example 1

- We want to find the true biological variance of gene g, that is we want to calculate $\frac{1}{C}\sum_{c=1}^{C}(X_{gc} - \bar{X}_{g\cdot})^2$.
- By using the result from SAVER framework, we will know that the formula of the true biological variance of gene g from the denoised data should be $\frac{1}{C}\left[\sum_{c=1}^{C}(\hat{X}_{gc} - \bar{\hat{X}}_{g\cdot})^2 + \sum_{c=1}^{C}\hat{v}_{gc}\right]$
- The red term is the sum of the variance across cells, which will be ignored if we directly use the variance from the denoised data directly.

$$E[\frac{1}{C}\sum_{c=1}^{C}(X_{gc}-\bar{X}_{g.})^2|Y,\Lambda=\hat{\Lambda}]$$

$$=E[\frac{1}{C}\sum_{c=1}^{C}((X_{gc}-\mu_{gc})-(\bar{X}_{g.}-\bar{\mu}_{g.})+\mu_{gc}-\bar{\mu}_{g.})^2|Y,\Lambda=\hat{\Lambda}]$$

where $\mu_{gc}=E[X_{gc}|Y,\Lambda=\hat{\Lambda}]$ and $\bar{\mu}_{g.}=E[\bar{X}_{g.}|Y,\Lambda=\hat{\Lambda}]$

$$=E[\frac{1}{C}\sum_{c=1}^{C}((\mu_{gc}-\bar{\mu}_{g.})-(\epsilon_{gc}-\bar{\epsilon}_{g.}))^2|Y,\Lambda=\hat{\Lambda}]$$

where $\epsilon_{gc}=X_{gc}-\mu_{gc}$ and $\bar{\epsilon}_{g.}=\bar{X}_{g.}-\bar{\mu}_{g.}$

$$=E[\frac{1}{C}\sum_{c=1}^{C}(\mu_{gc}-\bar{\mu}_{g.})^2+\frac{1}{C}\sum_{c=1}^{C}(\epsilon_{gc}-\bar{\epsilon}_{g.})^2|Y,\Lambda=\hat{\Lambda}]$$

since $\sum_{c=1}^{C}(\mu_{gc}-\bar{\mu}_{g.})(\epsilon_{gc}-\bar{\epsilon}_{g.})=0$

$$=\frac{1}{C}\sum_{c=1}^{C}(\mu_{gc}-\bar{\mu}_{g.})^2+E[\frac{1}{C}\sum_{c=1}^{C}(\epsilon_{gc}-\bar{\epsilon}_{g.})^2|Y,\Lambda=\hat{\Lambda}]$$

$$\approx\frac{1}{C}\sum_{c=1}^{C}(\hat{X}_{gc}-\bar{\hat{X}}_{g.})^2+\frac{1}{C}\sum_{c=1}^{C}v_{gc}$$

where v_{gc} is a posterior variance of gene g in cell c.

Estimating Function of X: Example 2

- We want to find the Pearson's correlation between two genes, g1 and g2, that is we want to calculate $\frac{\frac{1}{C}(X_{g1c}-\bar{X}_{g1\cdot})(X_{g2c}-\bar{X}_{g2\cdot})}{\sqrt{V_{g1}(X)\ V_{g2}(X)}}$, where $V_g(X)$ is a true biological variance of gene g.
- From the Example 1, we know that if we use the denoised data directly, we will underestimate the true biological variance.
- By using the result from SAVER framework and C is large enough, we will get that the correlation between two genes should be $\frac{\frac{1}{C}(\hat{X}_{g1c}-\bar{\hat{X}}_{g1\cdot})(\hat{X}_{g2c}-\bar{\hat{X}}_{g2\cdot})}{E[\sqrt{V_{g1}(X)}|Y,\Lambda=\hat{\Lambda}]\ E[\sqrt{V_{g2}(X)}|Y,\Lambda=\hat{\Lambda}]}$
  - The detailed derivation is in the supplementary material.

Estimating Function of X: Example 3

- This example will related to the dimension reduction and visualization.
- Visualization is a first step to analyze the single cell data. Dimension Reduction helps to construct the visualization since the single cell data is a high dimensional data.
- Euclidean distance is widely used for constructing the dissimilarity matrix (cell-to-cell distance).
- The Euclidean distance between two transcriptome profile is $d^2(c, c') = \sum_g (X_{gc} - X_{gc'})^2$
- By using the result from SAVER, the Euclidean distance is $\sum_g (\hat{X_{gc}} - \hat{X_{gc'}})^2 + \sum_g \hat{v}_{gc} + \sum_g \hat{v}_{gc'}$

$$E[d^2(c,c')|Y,\Lambda=\hat{\Lambda}] = E[\sum_g (X_{gc}-X_{gc'})^2|Y,\Lambda=\hat{\Lambda}]$$

$$= E[\sum_g ((X_{gc}-\hat{X}_{gc})-(X_{gc'}-\hat{X}_{gc'})+\hat{X}_{gc}-\hat{X}_{gc'})^2|Y,\Lambda=\hat{\Lambda}]$$

$$= E[\sum_g ((\hat{X}_{gc}-\hat{X}_{gc'})+(\epsilon_{gc}-\epsilon_{gc'}))^2|Y,\Lambda=\hat{\Lambda}]$$

where $\epsilon_{ab} = X_{ab} - \hat{X}_{ab}$

$$= E[\sum_g (\hat{X}_{gc}-\hat{X}_{gc'})^2|Y,\Lambda=\hat{\Lambda}]$$
$$+ \mathrm{E}[\sum_g (\epsilon_{gc}-\epsilon_{gc'})^2|Y,\Lambda=\hat{\Lambda}]$$
$$+ 2\mathrm{E}[\sum_g (\hat{X}_{gc}-\hat{X}_{gc'})(\epsilon_{gc}-\epsilon_{gc'})|Y,\Lambda=\hat{\Lambda}]$$

$$= \sum_g (\hat{X}_{gc}-\hat{X}_{gc'})^2 + E[\sum_g (\epsilon_{gc}-\epsilon_{gc'})^2|Y,\Lambda=\hat{\Lambda}]$$

since the cross product term is equal to 0.

$$= \sum_g (\hat{X}_{gc}-\hat{X}_{gc'})^2 + \sum_g (v_{gc}+v_{gc'})$$

where v_{gc} is a posterior variance, $Var[X_{gc}|Y_{gc},\Lambda=\hat{\Lambda}]$

Estimating Function of X: Example 3

Should these corrections improve the visualization?

- Consider $\sum_g (\hat{X_{gc}} - \hat{X_{gc'}})^2 + \sum_g \hat{v}_{gc} + \sum_g \hat{v}_{gc'}$
  - The distance calculated from the denoised matrix must be adjusted by adding the sums of the posterior variances across the genes.
  - If the red terms is much smaller than the first term, there should be a little effect on the visualization.

Estimating Function of X: Example 3

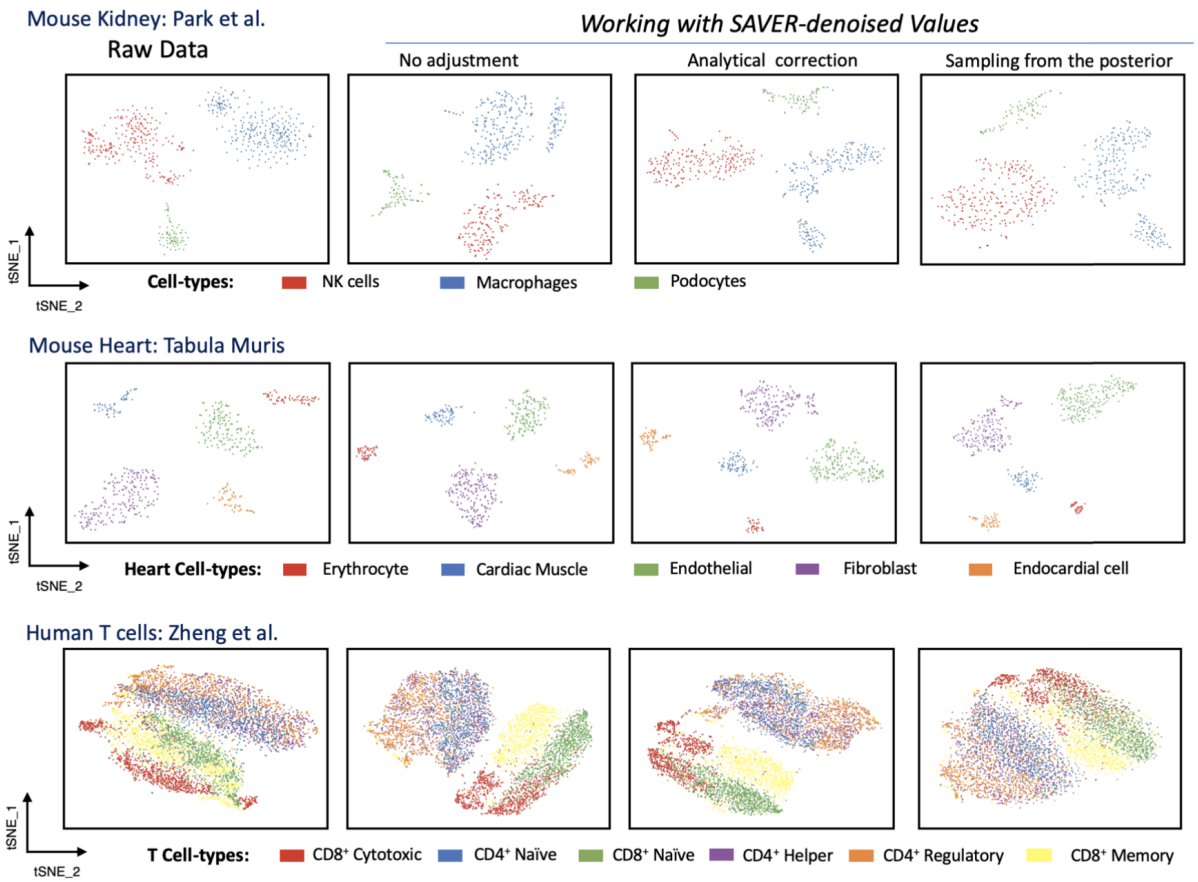

FIG. 5. *Visualizations obtained from three types of denoised distance matrices. tSNE visualizations of three immune cell types in the mouse kidney* (Park et al., 2018), *five major cell types that constitute the mouse heart* (The Tabula Muris Consortium, 2018), *and human T cells* (Zheng et al., 2017) *using distance matrices computed from raw and denoised data are shown.*

Estimating Function of X: Example 3

- Second row (Mouse Heart Data)
  - The data is still separated, same as the original.
  - The distance in the adjusted denoised plot did not represent the corrected distance.
- Third row (Human T. Cell Data)
  - This dataset is difficult since each group are quite similar to each other.
  - Analytic correction performed better than the sampling-based correction.

Estimating Function of X: Example 4

- This example will talk about clustering.
- The label resulted from clustering can be considered as a function of X.
- Clustering algorithms will use cell-to-cell distance. (Example 3)
- ARI Index (Adjusted Rand Index)
  - measuring the similarity of data points in the clusters.
  - the value of ARI is between -1 and 1.
  - the ARI should be high.
  - ARI $= 0$ means that everything is randomly assigned into the clusters.

Estimating Function of X: Example 4

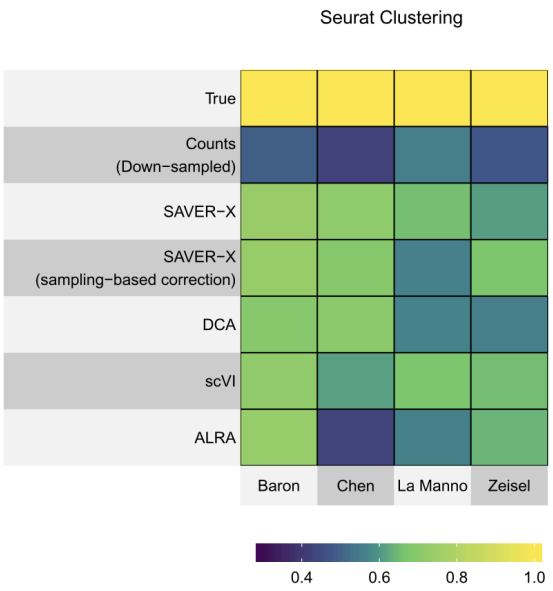

FIG. 6. *Effect on cell clustering. Heatmap of the adjusted rand index for Seurat clustering of the four data sets (Baron, Chen, La Manno and Zeisel) starting from the denoised values obtained from 7 methods (Original data treated as truth, down-sampled raw counts, SAVER-X, SAVER-X with posterior sampling, DCA, scVI, and ALRA).*

- The result from experiment shows that the denoising methods improved the clustering accuracy. However, the accuracy does not improve very much.

Gene Expression Differential Testing

- The common analysis of scRNA-seq
  - Finding genes that have different distributions between two groups of cells.
  - to test for differentially expressed genes (DE).
- Marker genes are genes that are highly expressed in only select cell type.
- Many denoising methods make the contrast of marker genes between cell clusters more intense. In the same time, the denoising methods tend to increase the false positive rate beyond the nominal value.

Silico Experiment

- The common strategy for accessing the false positive rate and power precisely is to create the artificial data sets through some combination of permutation, data splitting, subsampling, or simulation.
- Method used: Adapted from the third design proposed by Andrews and Hamberg (2018)
  - Select two cell types by using the labels provided by the original papers.
  - For each chosen cell type, it must have at least 100 cells.
  - Choose the similar cell types to create the difficult DE testing scenario.
- The result from this experiment is that all of Denoising method used in this experiment increased the false discovery proportion rate.

Adjusted Wilcoxon-rank sum test

- Wilcoxon-rank sum test is an alternative of two sample t-test.

Standard Wilcoxon-rank sum test

(The first group of cell) Assume that $X_{gi_1}, X_{gi_2}, ..., X_{gi_m} \overset{iid}{\sim} F_{g1}$
(The second group of cell) Assume that $X_{gj_1}, X_{gj_2}, ..., X_{gj_m} \overset{iid}{\sim} F_{g2}$
The null hypothesis $H_{0g} : F_{g1} = F_{g2}$
The test statistic is $U(X_{g.}) \sim N(0, 1)$

- Since we are not observe X and the test statistic will calculate from $\hat{X}$ instead, the type-I error will be inflated.

Adjusted Wilcoxon-rank sum test

- $U(X_{g\cdot})$ will be estimated by using the result from SAVER/SAVER-X. The result is $U(X_{g\cdot})|Y, \Lambda \sim N(\hat{\mu}_g, \hat{\sigma}_g^2)$
- We will rewrite the unobserved quantity, $U(X_{g\cdot})$ $U(X_{g\cdot}) \approx \hat{\mu}_g + \hat{\sigma}_g^2(Y, \Lambda)\epsilon$ where $\epsilon \sim N(0, 1)$
  - Derive the information of $\hat{\mu}_g$ under H_{0g}. $\hat{\mu}_g \sim N(0, \tau^2)$ where $\tau^2 = 2(1 + \hat{\sigma}_g^2)$
  - Since ϵ is independent of Y and Λ under H_{0g}, $Var[\hat{\mu}_g] \leq Var[U(X_g)] = 1$
- By using this method, type-I inflation is reduced but it still exceeded the nominal value.

Power in marker gene identification

- We already know that denoising emphasizes the contrast of the marker genes across the cell clusters.
- There are some questions that we still want to know.
  - Can we discover the new marker genes through a multiple hypothesis test?
  - Are known marker genes significant in the denoised data after the correction?
- Result from the experiment
  - Denoising data then using Adjusted Wilcoxon-rank sum test will allow us to detect all marker genes. Some of them will be insignificant in the raw data.

Reference

- Main Papers
  1. Data Denoising and Post-Denoising Corrections in Single Cell RNA Sequencing
     Agarwal D., Wang J. and Zhang N. R. (2020)
  2. Supplementary Materials for Data Denoising and Post-denoising Corrections in Single Cell RNA Sequencing
     Agarwal D., Wang J. and Zhang N. R. (2020)
- SAVER
  1. SAVER: gene expression recovery for single-cell RNA sequencing
     Huang, M., Wang, J., Torre, E. et al. (2018)